Intelligent Recommendation for Personalized Educational Intervention for Children with Autism

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***Abstract*—Educational intervention is one of the most important and effective treatment methods for children with autism spectrum disorder (ASD). However, limited educational resources are unable to meet the increasing needs of ASD patients. To this end, we propose a recommendation algorithm based on graph neural network (GNN), which intelligently recommends intervention activities and games for personalized needs of children with ASD. The proposed method characterizes ASD users according to their PEP-3 (Psycho educational Profile, PEP) scale evaluation data. Since the data on children's intervention behaviour contains multiple types of entity relationships, we construct graph model through meta-paths to mine deeper semantic associations and enable recommendations for intervention activities and games. The experimental results validate that our method perform well in all four recommended evaluation indicators, proving its effectiveness.**

Keywords—autism intervention, intelligent recommendation, graph neural network, intervention game recommendation

I. INTRODUCTION

Autism, also referred to as autism spectrum disorder (ASD), is a pervasive neurodevelopmental disorder[1].According to statistics from the US Centers for Disease Control and Prevention, the prevalence of autistic children has reached 1/36 in 2023 [2]. The latest survey data shows that the incidence rate of autism in our country has reached 0.79% [3]. This means that there are more than 10 million people with ASD in China, and the number is increasing by nearly 200,000 every year. Educational intervention methods are the primary means of rehabilitation for children with ASD. The formulation of intervention is mainly based on intervention activities and games. Intervention activities[4] involve rehabilitation therapists guiding and training children to complete some carefully designed learning activities. Intervention games[5] refer to the use of human-computer interaction to train children's various ability development through gamified resources. Several studies have shown that encouraging children with ASD to participate in developmentally appropriate games plays an important role in helping them learn to imitate, observe, demonstrate and ultimately develop social interaction skills.

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Despite educational intervention has achieved good results in children with ASD, it still faces the following problems:(1) Insufficient teachers engaged in educational intervention. According to relevant data from the China Disabled Persons' Federation, by the end of 2020, the number of rehabilitation institutions for children with ASD nationwide reached 2,681, serving less than 500,000 people, which was unable to meet the educational intervention needs. (2) The autism intervention industry in China started relatively late, with some policies and industry standards lacking. This leads to uneven qualifications of rehabilitation institutions and practitioners, as well as insufficient intervention resources. (3) The intervention cost for autism is high. This has increased the economic burden on families with ASD and also affected the frequency of rehabilitation training for a large proportion of children. (4) Individual intervention rely on professionals, with complex processes, heavy tasks, difficult implementation, and low effectiveness, resulting in insufficient intelligence level. In view of these problems, it is urgent to use intelligent approaches to generate individualized educational intervention for autistic children and provide corresponding intervention resources.

The essence of automatic autism intervention generation is to recommend personalized intervention activities and games based on children's core symptoms. Intelligent recommendation technology has been widely used in education, especially in online platforms like Massive Open Online Courses (MOOCs) and NetEase Cloud Classroom, which recommend new learning resources based on users' historical click data. However, its application in special education is lacking. The main reasons are: (1) Children with ASD have large individual differences and need to make targeted recommendations based on their developmental abilities. (2) There are fewer platforms related to special education and a lack of recommended datasets related to autism interventions.

In response to these problems, we propose an intelligent educational intervention recommendation method based on graph convolution network[6] and meta-path[7], which realizes personalized intervention resources for autistic patients, including intervention activities and games. The intervention activities are from "The Activity Guides for Pre-school Education"[8] authoritatively published by Hong Kong Heep Hong Society, and the intervention games are from the independently developed Star Future System[9]. The system is implemented by the recommendation algorithm based on graph

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neural network. Initially, it extracts content features from both users and projects. Subsequently, it constructs relevant meta-paths from the users' interaction logs and score data. These extracted content features and constructed meta-paths are fed into a graph convolutional network to derive the user and project feature representations. Ultimately, matrix factorization generates the final prediction matrix, enabling personalized recommendations of suitable intervention activities and games.

II. Autism Intervention Intelligent Recommendation System

The intelligent recommendation method harnesses AI technology to emulate how human experts devise interventions for autistic children based on their diagnostic assessments. More specifically, the PEP-3[10-11] assessment data of these children is fed into the system, which then employs its recommendation model to generate personalized intervention activities and games.

The intervention activities are from "The Activity Guides for Pre-school Education". The guide covers seven domains of cognitive activities, including sensory perception, causality, spatial concepts, etc. Each learning area has a corresponding learning focus. Taking sensory perception as an example, there are points: vision, hearing, and touch. Each learning focus has corresponding intervention items. Taking vision as an example, there are items such as responding to light, gazing, and pursuing. Finally, each intervention project has corresponding intervention goals. Taking visual recognition as an example, there are categories: recognizing faces of familiar people, recognizing common objects, and recognizing reduced physical models. In addition, each intervention target has a corresponding recommended age. For example, the recommended age for the recognition of faces of familiar people is 5-6 months.

This article uses game resources for the independently developed Star Future System. With ability intervention training for autistic children as its main function, the Star Future System aims to create an interactive intelligent learning environment where children can explore social interaction and cooperation skills in different social scenarios through human-computer interactive games, and practice multi-dimensional abilities such as cognitive comprehension, gross and fine motor skills in a safe and free environment. Based on the theory of psychological functions of autistic children, it provides autistic children with training materials and games in eight areas: social skills, cognitive understanding, classification concepts, memory enhancement, competitive learning, cooperative learning, fine motor and visual application. There are a variety of games. The game functions include pointing, dragging, drawing, pinching and other gesture operations as well as voice interaction, animation interaction and other methods to fully utilise the multi-dimensional abilities of autistic children. Figure 1 shows our Star Future system.



Fig. 1. Star Future system

Our intervention is a model that combines traditional intervention activities with human-computer interactive games. Both have advantages and limits, in which the intervention activities are more professional and have a better effect. Although it is the main tool used by rehabilitators, the children easily feel bored and rejected. While intervention games are fun-oriented, children with ASD are more likely to access electronic devices as they feel uncomfortable with human-to-human communication[12], but it still needs to improve generalisation skills in real-life scenarios. Introducing these two elements at the same time may have complementary effects.

III. Proposed Approach

In this article, we propose an algorithm that transforms autism data into a heterogeneous information network and realizes the recommendation of intervention activities and games using a graph neural network. The algorithm includes the following four steps: feature extraction, meta-path construction, graph convolution based on attention mechanism[13] and prediction matrix generation. The specific system architecture diagram is shown in Figure 2.

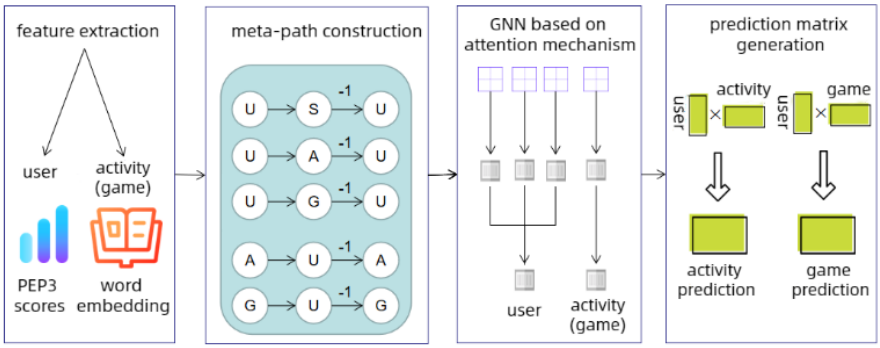


Fig. 2. The architecture of recommendation algorithm

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A. Feature Extraction

The input features of the graph neural network mainly include user features and project features. In terms of user characteristics, we use the evaluation score data of the PEP-3 as the user characteristics, represented by U. In terms of project characteristics, the name of a project is a high-level summary of the entire project content, which contains rich semantic information. Therefore, we use the Bert[14] word vectors of intervention activity names and game names as their respective features, represented by A and G respectively.

B. Metapath Construction

The heterogeneous network contains various entities such as users, intervention activities, intervention games, and complex relationships between entities. In order to express these relationships, this paper constructs multiple meta-paths. The meta path is defined as follows:

Definition: The meta-path is defined as $P_1 \xrightarrow{R_1} P_2 \xrightarrow{R_2} \ldots \xrightarrow{R_l} P_{l+1}$ which describes a composite relationship $R = R_1 \circ R_2 \circ \ldots \circ R$ between object P_1 and P_{l+1}, where $\circ$ represents a compound operation on a relation.

Two users can be connected through multiple meta-paths, such as user → intervention activity → user and user (UAU) → game → user (UGU). Different meta-paths reveal different semantics. The former indicates that two users have completed the same intervention activity, the latter indicates that two users have completed the same game.

Given that the more times two users complete the same activity, the more times they complete the same game, and the more similar the scale evaluation results are, the more similar their recommendation results should be. Accordingly, we construct meta-paths for intervention activity and game respectively. Among them, there are three user-based meta-paths, defined as A1, A2, and A3 respectively. A1 represents USU, that is, users with the same scale evaluation item greater than zero. A2 represents UAU, that is, users who have completed the same intervention activity. And A3 represents UGU, that is, users who have completed the same game. Intervention activities and games each have one meta-path, defined as B1 and B2 respectively. B1 represents AUA, which are two intervention activities completed by the same user, while B2 represents GUG, which is two games completed by the same user.

It can be seen that the user meta-path between intervention activity and intervention game recommendation is the same, and the difference lies in the item meta-path. According to the designed meta-path, the corresponding adjacency matrix can be obtained. Taking UAU as an example, $M=W_{UA} \bullet W_{AU}$, M represents the adjacency matrix obtained by constructing the meta-path, W_{UA} represents the record of the user completing the intervention activity, and W_{AU} represents the transpose of W_{UA}.

C. Graph Convolutional Neural Network Based on Attention Mechanism

After obtaining user features, item features, user adjacency matrix and item adjacency matrix, they can be all fed into a graph convolutional network to learn potential entity representation of users, intervention activities and games. Among them, three graph convolutional neural networks are constructed based on the given three user element paths. The user features are respectively processed by the graph convolutional neural network and then fused through the attention mechanism to obtain the final user entity representation. For the entity representation of intervention activities and games, a graph convolutional neural network is constructed based on AUA and GUG respectively. The graph neural network will convert the corresponding word vector into an entity representation for recommendation.

Specifically, we adopt a multiple-layer graph convolutional network (GCN) with layer-wise propagation rule:

$$H^{(l+1)}=\sigma(\widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}H^{l}W^{l}) \quad (1)$$

where $h^{(l+1)}$ represents the feature vector matrix of the entity in the l+1th layer, and l represents the number of layers of graph convolution. In particular, h^0 represents the feature vector matrix of the original entity in this article, that is, the input user feature or item feature. σ is a nonlinear activation function, and RELU activation is used in this article. A is the adjacency matrix, that is, the relationship matrix obtained by constructing the element path. $\widetilde{A}=A+I$ is the result of the self-connection of the adjacency matrix. I is the identity matrix. $\widetilde{D}$ is the degree matrix of $\widetilde{A}$. W^l represents the current layer volume. The trainable parameter matrix of the product transformation. This article uses e to represent the entity. If the corresponding meta-path is M, then $e_M=H^3$. There is only one meta-path for intervention activities and games respectively. For users, there are three meta-paths, namely A_1, A_2, A_3. The multiple calculated entities need to be fused, that is, by setting the weights, the three results are summed to obtain the final result. In order to solve this problem, this paper introduces the attention mechanism. The merging of multiple meta-paths is represented as follows:

$$e=\sum_{i=1}^{|M|} \alpha_{M_i} e_{M_i} \quad (2)$$

Among them, e represents the result of entity fusion, e_{M_i} represents the entity representation of several meta-paths that need to be fused, and $|M|$ represents the number of meta-paths that need to be fused. α_{M_i} represents the weight of each meta-path. The weight calculation is represented as follows:

$$\alpha_{M_i} = \frac{\exp(\sigma(a e_{M_i}))}{\sum j \in |M| \exp(\sigma(a e_{M_j}))} \quad (3)$$

Among them, e_{M_i} is the entity representation of the target meta-path, and e_{M_j} is the entity representation of other meta-paths, and a represents a trainable attention vector, and α represents the nonlinear activation function, and the softmax function is used here to Attention weights are normalized.

D. Prediction Matrix Generation

Through graph convolution network based on the attention mechanism, the final representation of the user concept e^u, the final representation of the intervention activity concept e^a and the final representation of the game concept e^g have been obtained. Next, the matrix factorization (MF)[15] method is used

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to recommend users Intervention activities or games. Here, the number of times the user completes intervention activities and games is used as the demand matrix for intervention activity recommendation and game recommendation respectively, and the entity representation of users, intervention activities and games is introduced, as shown in (4-5):

$$\widehat{d_{u,a}} = x_u^T y_a + \beta_u \cdot e^{uT} t^a + \beta_a \cdot t^{uT} e^a \quad (4)$$

$$\widehat{d_{u,g}} = x_u^T y_g + \beta_u \cdot e^{uT} t^g + \beta_g \cdot t^{uT} e^g \quad (5)$$

Among them, e^u, e^a, and e^g respectively represent the entity representation of the user, intervention activity concept, and game concept; the trainable parameters tu, ta, tg are introduced to ensure that e^u, e^a and e^u, e^g are in the same space. And β^u, β^a, β^g are an adjustment parameter, the purpose is to achieve a suitable demand forecast. After introducing the regularization term, the objective function is as shown in (6-7):

$$\min_{u,a} \frac{1}{j \times k} \sum_{u=1}^{k} \sum_{a=1}^{j} \left(d_{u,a} - \hat{d}_{u,a}\right)^2 + \lambda\left(\left||x_u|\right|_2 + \left||y_a|\right|_2 + \left||t^u|\right|_2 + \left||t^a|\right|_2\right) \quad (6)$$

$$\min_{u,a} \frac{1}{j \times l} \sum_{u=1}^{l} \sum_{g=1}^{j} \left(d_{u,g} - \hat{d}_{u,g}\right)^2 + \lambda\left(\left||x_u|\right|_2 + \left||y_g|\right|_2 + \left||t^u|\right|_2 + \left||t^g|\right|_2\right) \quad (7)$$

Here λ is the regularization parameter, and the stochastic gradient descent algorithm can be used to optimize the final objective function.

IV. Experiments and Analysis

To validate the efficacy of the proposed method, we construct a dataset specifically for recommending intervention activities and games. Subsequently, it evaluates the algorithms' performance using commonly adopted metrics like HR, NDCG, MRR, and AUC, culminating in an analysis of the experimental results.

A. Dataset

In this experiment, a total of 27 children diagnosed with ASD were recruited as subjects in a pediatric hospital. Before the experiment, informed consent forms were obtained from their respective guardians. The children, aged 18-74 months, fulfilled the diagnostic criteria of the Diagnostic and Statistical Manual of Mental Disorders (DSM), Fifth Edition, United States of America, and their diagnosis was confirmed in a double-blind manner by two deputy directors of the Department of Developmental and Behavioural Paediatrics. Children with schizophrenia, epilepsy and other neurological disorders were excluded. Normal visual and auditory development of the subject children was determined prior to the experiment.

In order to build user intervention activity recommendation data, this article follows the "The Activity Guides for Pre-school Education" to make manual recommendations to each user as training data for the model. The guide lists a total of 347 cognitive intervention activities, and each activity has a corresponding recommended age. We invited two rehabilitation therapists to recommend appropriate intervention activities from 347 activities for these 27 children to formulate intervention. Each child received approximately 10-50 intervention activities. According to the intervention, the last recommended activities are used as test data, and the remaining data are used as training data.

In order to construct user intervention game recommendation data, we also invited a number of special education experts to annotate 212 games in the Star Future system, including the game's intervention field, intervention age and personalized preferences. Then, two rehabilitation therapists were invited to develop game intervention for the 27 children based on their PEP-3 assessment results and game labels. The number of recommended games for each child was approximately 10-50. Training and test data are generated in the same way as intervention campaign data.

In order to better train the graph neural network, this article uses the following two methods to enhance the user's training data. (1) Randomly select one dimension of the user's 34-dimensional features for perturbation; (2) Randomly replace a recommended item. After the data was enhanced 200 times, a total of 27*2*200=10800 pieces of data were obtained.

B. Evaluation Metrics and Experiments Results

To evaluate recommendation performance, we pair each target instance in the test set with 99 randomly sampled negative instances and output the prediction scores for 100 instances. This article uses evaluation indicators commonly used in recommendation algorithms, including the hit rate of the top K items (HR@K), the normalized loss cumulative return of the top K items (NDCG@K), the average reciprocal ranking (MRR) and the receiver operation. Area under the characteristic curve (ROC) (AUC). This article sets K to 5, 10, and 20 respectively, and calculates all indicators for every 100 instances (1 positive instance plus 99 negative instances). The evaluation results are shown in Table I below.

It can be seen from the table I that the intervention activity recommendation and game recommendation have achieved good results. For the measurement of HR, take HR@10 as an example. Among the 10 projects recommended in this article, the hit probability reaches 95% and 97% respectively. For the measurement of NDCG, take NDCG@10 as an example. Among the 10 projects recommended in this article, the similarity with the target list reaches 80% and 88% respectively. The MRR indicators reached 0.75 and 0.86 respectively, which means that the hit positions recommended in this article are quite high. The AUC indicators reached 0.97 and 0.98 respectively, which means that the probability that the positive instances recommended in this article are ranked in front of the negative instances are 97% and 98% respectively.

Among them, the results of game recommendation are slightly better than those of activity recommendation, because the number of game items is less than that of activity items. The results recommended by the two rehabilitation therapists are more consistent, and the difficulty of game recommendation is relatively less. Overall, this intelligent recommendation can generally meet the needs of intelligent intervention for autism.

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TABLE I. RESULTS WITH ACTIVITY RECOMMENDATION AND GAME RECOMMENDATION

Metrics	HR@5	HR@10	HR@20	nDCG@5	nDCG@10	nDCG@20	MRR	AUC
Activity	0.90	0.97	0.99	0.81	0.83	0.84	0.80	0.98
Game	0.93	0.97	0.99	0.87	0.88	0.89	0.86	0.98

In addition, according to different meta-paths, we conducted comparison experiments with different combinations to investigate the impact of the type and number of meta-paths on the results. The comparison metrics used are HR@5, nDCG@5, MRR and AUC, and the metrics are shown in Table II .

TABLE II. DIFFERENT META-PATH RESULTS

metapaths	HR@5	nDCG@5	MRR	AUC
UAU	0.86	0.79	0.78	0.95
UAU&UGU	0.88	0.81	0.79	0.97
UAU&USU	0.89	0.81	0.79	0.98
UAU&UGU&USU	0.90	0.81	0.80	0.98
UGU	0.81	0.73	0.72	0.96
UGU&UAU	0.87	0.81	0.80	0.97
UGU&USU	0.82	0.74	0.73	0.96
UGU&UAU&USU	0.93	0.87	0.86	0.98

The overall results show that more combinations of meta-paths show better performance. For activity recommendation, the impact of meta-path combination change is relatively small, and the importance of USU is slightly larger than that of UGU. for game recommendation, the impact of meta-path combination change is larger, and the importance of UAU is larger than that of USU.

V. CONCLUSION

In this work, we propose a personalized recommendation method for intervention activities and intervention games based on the developmental abilities of ASD. Specifically, we propose an intelligent recommendation system grounded in PEP-3 scale assessments, which designs and supplies tailored intervention activities and game resources for children of varying abilities. Furthermore, a graph neural network algorithm leveraging multiple meta-paths constructs a heterogeneous information network among user's scores on the PEP-3 Assessment, intervention activities and intervention games. This approach employs graph neural networks to deliver intelligent recommendations. Additionally, a dataset dedicated to recommending autism intervention activities and games has been established, achieving an AUC score of 0.98. Notably, the results show a strong consistency with expert-recommended solutions, thus providing a robust data foundation for further practical application.

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